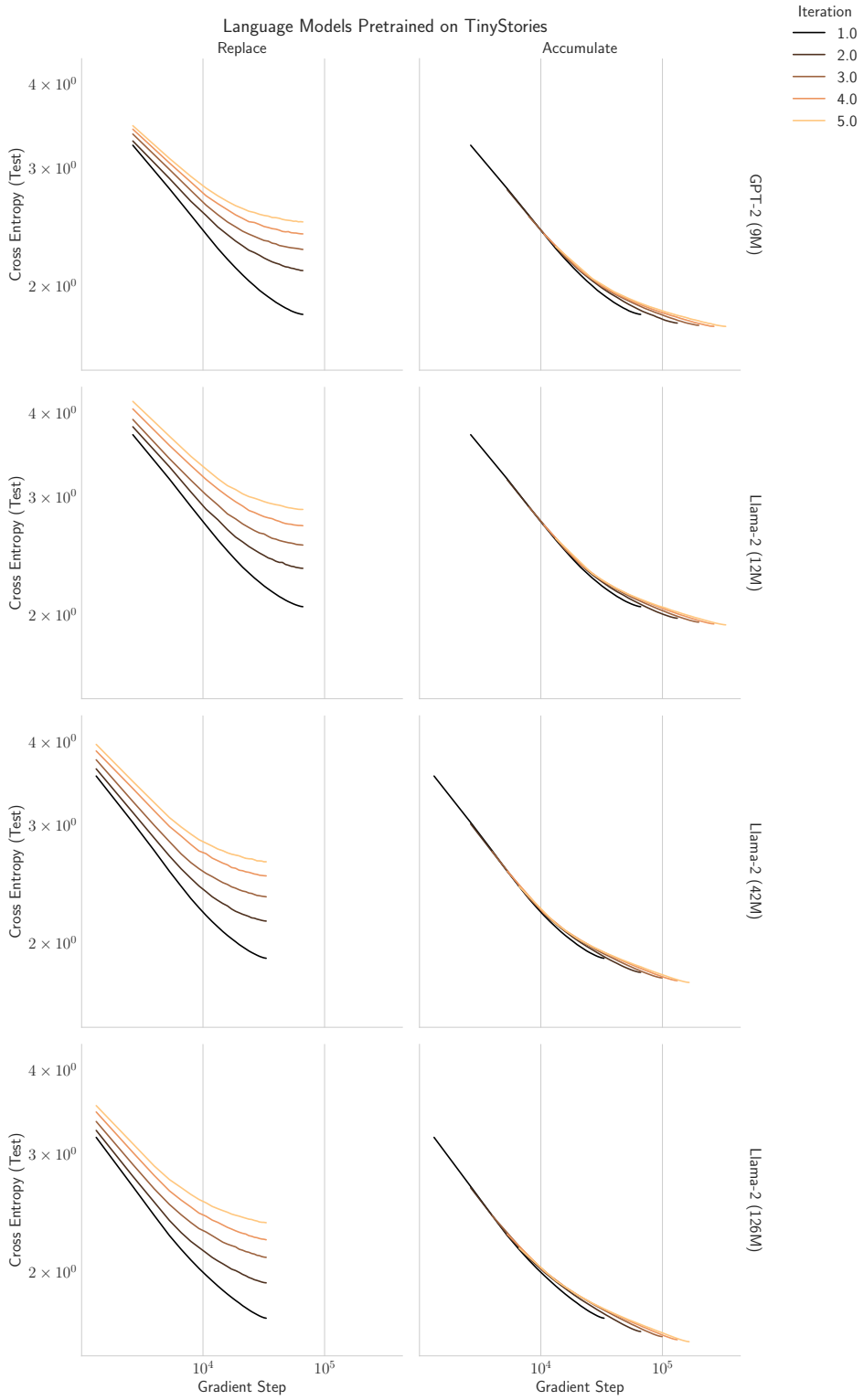

Figure 12: **Data Accumulation Avoids Model Collapse in Language Modeling.** Learning curves for individual model-fitting iterations when repeatedly *replacing* data (left), and when *accumulating* data (right), in log-log scale.

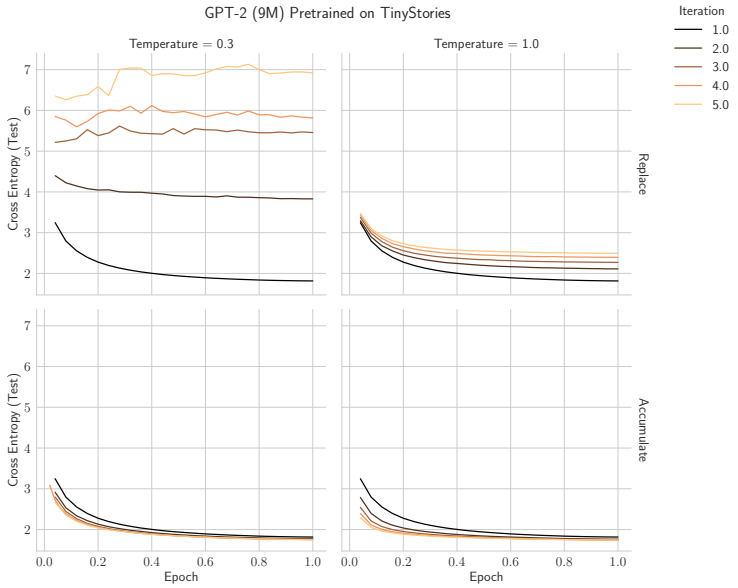

Figure 13: Accumulating data shows stable behavior across different generation temperatures for a GPT-2 (9M) model, while replacing data does not.

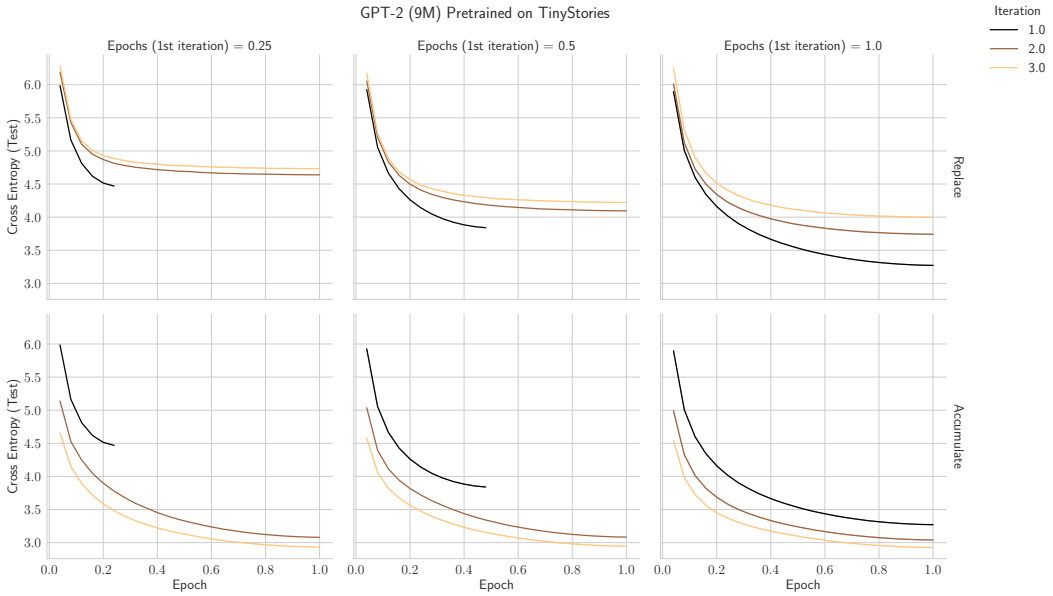

Figure 14: Model quality after the first model-fitting iteration does not qualitatively change behavior in subsequent iterations. Columns show differing training amount (as measure by epochs) in first iteration.

D Additional Details on VAE Experiments

Experiment Details. As pre-processing, we crop and down-sample the images to 64x64 pixels. We use a standard convolutional architecture for the VAE model consisting of 5 convolutional layers with 32, 64, 128, 256, and 512 channels, respectively, and a similar convolutional decoder structure. The latent space is 128-dimensional isotropic Gaussian, represented by 2 MLP layers. Each data iteration consists of 100 training epochs, after which we generate 163K new training images by sampling latents from the Gaussian prior and the passing them through the generator model.

Analysis of Reconstructions. Figure 15 shows reconstructions after replacing (left) and accumulating (center) data, compared to baseline (right). Analyzing the reconstruction of test set images also reveals interesting findings - the model trained only on data from the prior iteration has indeed collapsed and cannot represent any other classes besides the single mode it generates. Interestingly, the model trained on aggregated data still maintains it's capabilities and generates accurate reconstructions, including smaller details such as glasses and hats. We hypothesize that this model maintains it's generative capabilities, but these details become a more minor sub-manifold in the latent space, which is realigned with the newly-generated data, hence why they appear less often in the generated images, which use samples from the prior.

Figure 15: **Data Accumulation Maintains Model Capabilities.** Image reconstructions from the test set. Left: Training on prior iterations collapses the model's capability, and subsequently, it can only represent a single mode. Middle: training on aggregated data preserves model capabilities and leads to little to no degradation in the reconstructed images. Right: Baseline reconstructions after 100 training epochs on the dataset.